

Your grade: 100%Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.[Next item →](#)

1. If you have 10,000 examples, how would you split the train/dev/test set? Choose the best option.

1 / 1 point

- ☒ 60% train. 20% dev. 20% test.
- ☐ 33% train. 33% dev. 33% test.
- ☐ 98% train. 1% dev. 1% test.

 **Nice work**

Yes. This might be considered a small data set, not in the range of big data. Thus a more classical (old) best practice should be used.

2. In a personal experiment, an M.L. student decides to not use a test set, only train-dev sets. In this case which of the following is true?

1 / 1 point

- ☐ He won't be able to measure the variance of the model.
- ☐ Not having a test set is unacceptable under any circumstance.
- ☐ He won't be able to measure the bias of the model.
- ☒ He might be overfitting to the dev set.

 **Nice work**

Yes. Although not recommended, if a more accurate measure of the performance is not necessary it is ok to not use a test set. However, this might cause an overfit to the dev set.

3. If your Neural Network model seems to have high bias, what of the following would be promising things to try? (Check all that apply.)

1 / 1 point

- ☐ Get more training data
- ☒ Make the Neural Network deeper

 Nice work

- ☒ Increase the number of units in each hidden layer

 Nice work

- ☐ Add regularization

4. You are working on an automated check-out kiosk for a supermarket and are building a classifier for apples, bananas, and oranges.

1 / 1 point

Suppose your classifier obtains a training set error of 19% and a development set error of 21%.

Which of the following is the most promising strategy to improve your classifier? (Assume the human error is approximately 0%)

- ☐ Increase the regularization parameter lambda.
- ☐ Get more training data.
- ☒ Use a bigger network.

 Nice work

A larger network can reduce bias by enabling the model to learn more complex patterns.

5. Which of the following are regularization techniques?

1 / 1 point

- ☐ Gradient Checking.
- ☐ Increase the number of layers of the network.
- ☒ Weight decay.

✔ Nice work

Correct. Weight decay is a form of regularization.

- ☒ Dropout.

✔ Nice work

Correct. Using dropout layers is a regularization technique.

6. **True or False:** To reduce high variance, the regularization hyperparameter lambda must be increased.

1 / 1 point

- ☒ True
- ☐ False

✔ Nice work

Correct. By increasing the regularization parameter the magnitude of the weight parameters is reduced. This helps reduce the variance.

7. Which of the following are true about dropout?

1 / 1 point

7. Which of the following are true about dropout?

1 / 1 point

☒ It helps to reduce overfitting. **Nice work**

Correct. The dropout is a regularization technique and thus helps to reduce the overfit.

☐ It helps to reduce the bias of a model.☐ In practice, it eliminates units of each layer with a probability of keep_prob.☒ In practice, it eliminates units of each layer with a probability of 1- keep_prob. **Nice work**

Correct. The probability that dropout doesn't eliminate a neuron is keep_prob.

8. Increasing the parameter keep_prob from (say) 0.5 to 0.6 will likely cause the following: (Check the two that apply)

1 / 1 point

☐ Increasing the regularization effect☒ Reducing the regularization effect **Nice work**☐ Causing the neural network to end up with a higher training set error☒ Causing the neural network to end up with a lower training set error **Nice work**

9. Which of the following actions increase the regularization of a model? (Check all that apply)

1 / 1 point

- ☐ Decrease the value of the hyperparameter lambda.
- ☐ Use Xavier initialization.
- ☒ Increase the value of the hyperparameter lambda.

✔ Nice work

Correct. When increasing the hyperparameter lambda, we increase the effect of the L_2 penalization.

- ☒ Decrease the value of keep_prob in dropout.

✔ Nice work

Correct. When decreasing the keep_prob value, the probability that a node gets discarded during training is higher, thus reducing the regularization effect.

- ☐ Increase the value of keep_prob in dropout.

10. Which of the following is the correct expression to normalize the input $\mathbf{x}$?

1 / 1 point

- ☐ $x = \frac{x}{\sigma}$
- ☐ $x = \frac{1}{m} \sum_{i=1}^m (x^{(i)})^2$
- ☒ $x = \frac{x - \mu}{\sigma}$
- ☐ $x = \frac{1}{m} \sum_{i=1}^m x^{(i)}$

✔ Nice work

Correct. This shifts the mean of the input to the origin and makes the variance one in each coordinate of the input examples.